NVIDIA × ARM ACQUISITION CASE STUDY (2026)

A STRATEGIC, FINANCIAL, AND REGULATORY ASSESSMENT

Prepared by: Akash Ghosal

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(This document is an independent analytical study based on publicly available information.)

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1. EXECUTIVE SUMMARY

This case study evaluates whether NVIDIA should reconsider acquiring Arm Holdings in the 2026 AI compute landscape and independently assesses Arm's intrinsic valuation using a scenario-based DCF model (Base, Bull, and Bear). The analysis combines industry dynamics, company fundamentals, and multi-scenario valuation outcomes to determine the attractiveness and feasibility of such a transaction.
The global semiconductor market is undergoing its most significant structural shift in decades. AI workloads, hyperscale data-center deployments, edge compute proliferation, and geopolitical realignments have dramatically increased the strategic importance of CPU and GPU architectures. NVIDIA currently dominates accelerated computing through CUDA and its data-center GPU ecosystem, while Arm powers 99% of the mobile CPU market and is expanding across cloud, automotive, and IoT. The long-term convergence of CPU–GPU architectures positions Arm as a foundational layer of next-generation AI infrastructure.
NVIDIA previously attempted to acquire Arm in 2020 for USD 40 billion, but the transaction was blocked in 2022 due to regulatory concerns around vertical foreclosure, industry neutrality, and global dependency on Arm's instruction set architecture (ISA). Since then, the market has shifted substantially. The AI boom has increased the strategic value of CPUs in heterogeneous compute, while new industry collaborations and geopolitical pressures have reshaped regulatory incentives. Although a renewed acquisition attempt would still face scrutiny, the context in 2026 differs meaningfully from 2020–2022.
A detailed standalone valuation of Arm was performed using a multi-scenario DCF model built on revenue forecasts, margin evolution, cash-flow projections, and sensitivity analysis.

- **Base Case Enterprise Value (EV): USD ~25.6 billion**
- **Bull Case EV: USD ~40–60 billion** (higher growth, margin expansion, lower WACC)
- **Bear Case EV: USD ~12–18 billion** (slower adoption, muted margins, higher discount rates)

Terminal value accounts for the majority of EV, reflecting Arm's long-duration licensing and royalty economics. Sensitivity analysis shows valuation is most responsive to WACC and terminal growth, underscoring the importance of long-term royalty scalability.
Strategically, NVIDIA would benefit from integrating Arm's CPU roadmap into its accelerated computing stack, strengthening leadership across data centers, automotive, robotics, and edge devices. However, any transaction would require stringent behavioural commitments, firewalls, or structural remedies to preserve industry neutrality. Even under an optimistic regulatory environment, an acquisition remains challenging but not impossible.
Based on the financial valuation, strategic upside, and regulatory constraints, the final view is that **Arm remains a high-value strategic asset, and while NVIDIA would benefit materially from acquiring Arm, the deal's feasibility depends heavily on global political alignment and enforceable neutrality commitments.** The DCF analysis reinforces that Arm's long-term value is driven by structural growth in AI compute, and NVIDIA stands to benefit whether as an owner or as a key ecosystem partner.

2. INTRODUCTION & OBJECTIVE OF THE CASE STUDY

The rapid evolution of artificial intelligence and accelerated computing has fundamentally reshaped the strategic importance of semiconductor architecture. As AI workloads increasingly dominate data-center, cloud, automotive, and edge applications, control over foundational compute layers—CPU, GPU, and system-on-chip (SoC) architectures—has become a decisive competitive advantage. This case study examines whether Arm Holdings, as a central pillar of global CPU architecture, represents a strategic and financial opportunity for NVIDIA in the contemporary AI-driven market landscape.

Arm occupies a unique position in the semiconductor ecosystem. Its licensing-based business model underpins the majority of the world's mobile processors and is rapidly expanding into cloud infrastructure, automotive systems, and IoT devices. Unlike traditional semiconductor firms, Arm operates as an architecture provider rather than a chip manufacturer, maintaining neutrality across an extensive ecosystem of licensees. This neutrality has historically been critical to its success, while simultaneously making ownership of Arm politically and regulatorily complex.

NVIDIA's attempted acquisition of Arm between 2020 and 2022 highlighted the strategic tension between vertical integration and ecosystem neutrality. Although the transaction was ultimately blocked due to global regulatory opposition, the rationale behind the acquisition reflected a long-term vision of converging CPU and GPU architectures to dominate future AI compute platforms. Since the termination of that deal, the semiconductor market has undergone a paradigm shift driven by explosive AI demand, heightened geopolitical competition, and accelerating compute specialization.

The primary objective of this case study is threefold.

First, it seeks to assess Arm's intrinsic value as a standalone entity using a structured, scenario-based valuation framework. This includes a detailed discounted cash flow (DCF) analysis incorporating Base, Bull, and Bear scenarios to capture uncertainty in growth, margins, and capital costs.

Second, the study evaluates whether the strategic logic that motivated NVIDIA's original acquisition attempt remains relevant—or has strengthened—in the 2026 market environment.

Third, it analyses whether evolving industry dynamics and regulatory conditions could materially alter the feasibility of such a transaction if reconsidered.

This case study is designed as a decision-oriented strategic and financial assessment rather than a transactional recommendation. It does not assume an acquisition will occur but instead uses the NVIDIA–Arm relationship as a lens to examine broader questions around value creation, competitive positioning, and regulatory constraints in the global semiconductor industry. By integrating financial modelling, strategic analysis, and regulatory considerations, the study aims to provide a balanced view of both opportunity and limitation in one of the most consequential potential transactions in modern semiconductor history.

3. GLOBAL SEMICONDUCTOR & AI COMPUTE INDUSTRY OVERVIEW

3.1 Global Semiconductor Value Chain

The global semiconductor industry operates through a highly specialized and fragmented value chain, where design, manufacturing, and distribution are dominated by different players across regions. At a high level, the value chain consists of semiconductor design (fabless companies), fabrication (foundries), assembly and testing, and downstream system integration.

Fabless firms such as NVIDIA, AMD, Qualcomm, and Arm focus on architecture design, intellectual property (IP) development, and software ecosystems, while manufacturing is outsourced to advanced foundries such as TSMC and Samsung. This structural separation has enabled rapid innovation but has also concentrated manufacturing risk in a small number of geographies. The increasing complexity and cost of advanced nodes (5nm, 3nm, and below) further reinforce the strategic importance of design-led firms that control critical compute architectures without bearing fabrication risk directly.

Within this ecosystem, Arm occupies a distinct role as an IP licensor rather than a chip producer. Its instruction set architecture (ISA) and CPU designs form a foundational layer upon which thousands of semiconductor products are built, positioning Arm as a systemic enabler rather than a traditional competitive manufacturer.

3.2 AI Compute Stack (CPU, GPU, NPU, DPU)

AI workloads have fundamentally altered the structure of compute demand. Traditional CPU-centric architectures are increasingly insufficient for training and inference tasks, leading to the emergence of heterogeneous compute stacks combining CPUs, GPUs, and specialized accelerators.

- **CPUs** remain central to orchestration, general-purpose workloads, and system-level coordination.
- **GPUs**, led overwhelmingly by NVIDIA, dominate parallel compute for AI training and large-scale inference.
- **NPUs (Neural Processing Units)** and **DPUs (Data Processing Units)** address specialized workloads such as low-latency inference, networking, and data movement.

The strategic importance of CPUs has not diminished; rather, it has evolved. Modern AI systems rely on tight CPU–GPU integration to optimize performance, power efficiency, and data throughput. Arm's architecture has gained traction in data centers due to its power efficiency, scalability, and growing software support, making it increasingly relevant beyond mobile devices. This shift strengthens the strategic case for alignment between CPU architecture providers like Arm and GPU-centric firms such as NVIDIA.

3.3 Key Players & Competitive Dynamics

The competitive landscape in AI compute is defined by ecosystem control rather than isolated hardware performance. NVIDIA dominates accelerated computing through its combined hardware and software moat, particularly its CUDA platform and vertically integrated data-center solutions. AMD competes across CPUs and GPUs with an open ecosystem approach, while Intel seeks to reassert relevance through manufacturing leadership and heterogeneous computing initiatives.

Arm competes indirectly with x86 architectures (Intel, AMD) and emerging open-source alternatives such as RISC-V. Its success is rooted in ecosystem neutrality, licensing flexibility, and long-term partnerships across technology firms that would otherwise be competitors. This neutrality has historically enabled Arm to achieve near-universal adoption in mobile and growing penetration in cloud and automotive applications.

As AI workloads scale, the balance of power increasingly depends on who controls key architectural standards and software compatibility layers. This dynamic elevates Arm from a background IP provider to a strategically critical asset within the global compute stack.

3.4 Global Supply Chain Risk & Geopolitical Exposure

The semiconductor industry is highly exposed to geopolitical risk due to geographic concentration in design leadership, fabrication capability, and raw material sourcing. Advanced semiconductor manufacturing is overwhelmingly concentrated in East Asia, with Taiwan representing the single most critical node in the global supply chain. As a result, governments have increasingly treated semiconductors as strategic infrastructure rather than purely commercial products.

This shift has led to policy interventions such as export controls, onshoring incentives, and national semiconductor strategies in the United States, Europe, and China. Arm's UK origins, global licensing model, and deep integration into both Western and Asian technology ecosystems make it particularly sensitive to geopolitical considerations. Any change in its ownership structure directly raises questions of national security, competitive neutrality, and supply chain resilience.

Against this backdrop, large-scale semiconductor transactions—especially those involving foundational IP—are no longer evaluated solely on economic grounds. Regulatory authorities increasingly assess their systemic importance, competitive effects, and long-term implications for innovation and national sovereignty.

4. COMPANY ANALYSIS: NVIDIA CORPORATION

4.1 Business Segments

NVIDIA operates as a global leader in accelerated computing, with its business structured around high-performance hardware platforms tightly integrated with proprietary software ecosystems. The company's core revenue is driven by its **Data Center** segment, which has emerged as the dominant contributor due to explosive demand for AI training and inference workloads. Additional segments include **Gaming**, **Professional Visualization**, **Automotive**, and **OEM & Other**, each addressing different end markets but sharing a common architectural foundation.

What differentiates NVIDIA's business model is not merely hardware performance, but its ability to embed its products deeply into customers' computational workflows. This integration creates high switching costs and reinforces long-term demand across cycles, particularly in enterprise and hyperscale environments.

4.2 Product Ecosystem (CUDA, GPUs, Data Centers)

At the center of NVIDIA's competitive advantage is **CUDA**, its proprietary parallel computing platform and software ecosystem. CUDA transforms NVIDIA's GPUs from commodity hardware into essential infrastructure for AI research, model training, and deployment. Over time, CUDA has evolved into a platform-level moat, with extensive developer adoption, optimized libraries, and deep compatibility across AI frameworks.

NVIDIA's GPU portfolio, particularly in data centers, underpins large-scale AI training models and high-performance inference applications. Beyond GPUs, NVIDIA has expanded into system-level solutions, including networking, interconnects, and AI-optimized server architectures. This full-stack approach positions NVIDIA not simply as a chip vendor, but as an AI infrastructure provider.

The growing convergence of CPUs and GPUs in heterogeneous computing environments increases the strategic importance of CPU architectures that can tightly integrate with NVIDIA's accelerated compute stack—an area where Arm's relevance becomes increasingly pronounced.

4.3 Financial Performance (Five-Year Overview)

Over the past five years, NVIDIA has exhibited exceptional financial performance characterized by strong revenue growth, expanding margins, and robust cash generation. The company's transition toward data center–driven growth has materially improved its revenue mix, reducing cyclicality associated with consumer markets such as gaming.

Operating margins have expanded significantly as software, platform-level economics, and scale efficiencies increasingly contribute to profitability. Free cash flow generation has strengthened correspondingly, providing NVIDIA with strategic flexibility in capital allocation, investments, and potential acquisitions. This financial strength enables NVIDIA to pursue long-duration strategic initiatives that would be inaccessible to less capitalized competitors.

4.4 Growth Drivers & Strategic Positioning

NVIDIA's growth strategy is centred on sustaining leadership in accelerated computing while expanding its footprint across new AI-driven applications. Key growth drivers include continued hyperscale investment in AI infrastructure, enterprise adoption of AI workflows, and expanding use cases in automotive autonomy, robotics, and edge computing.

Strategically, NVIDIA is positioned as the central enabler of AI compute rather than a peripheral hardware supplier. Its ecosystem-first approach allows the firm to benefit from both hardware demand and long-term software monetization. However, the sustainability of this leadership increasingly depends on control—or close alignment—with complementary compute architectures, particularly CPUs, that define system-level performance and power efficiency.

5. COMPANY ANALYSIS: ARM HOLDINGS

5.1 Business Model (Licensing & Royalties)

Arm operates under a fundamentally different business model from traditional semiconductor companies. Rather than manufacturing chips, Arm designs CPU architectures and licenses its intellectual property to a broad ecosystem of semiconductor companies. Revenue is generated through upfront licensing fees and long-term royalties linked to chip shipments, creating a capital-light, scalable business model with high gross margins.

This licensing framework allows Arm to benefit from industry-wide growth while maintaining neutrality across competitors. The royalty-based structure also results in highly predictable, long-duration cash flows, particularly as Arm's architectures become embedded in foundational technology platforms across multiple end markets.

5.2 Architecture Ecosystem & Market Penetration

Arm's architecture underpins approximately 99% of global smartphone processors and has achieved dominant penetration in mobile and embedded systems. More recently, Arm has expanded into cloud infrastructure, automotive compute, and edge devices, leveraging its power efficiency and scalability advantages.

The success of Arm's ecosystem is driven by its broad developer adoption, compatibility across operating systems, and long-standing relationships with semiconductor firms. Unlike vertically integrated competitors, Arm's value lies in enabling innovation across the industry rather than capturing it directly. This ecosystem-centric role makes Arm strategically indispensable but structurally sensitive to changes in ownership or control.

5.3 Financial Performance Post-IPO

Following its public listing, Arm has demonstrated stable revenue growth supported by expanding royalty streams and increasing penetration into high-performance computing markets. Investment in research and development remains substantial, reflecting the need to continuously advance architectural performance and maintain relevance across evolving workloads.

Despite near-term volatility associated with market cycles, Arm's long-term financial profile remains attractive due to operating leverage inherent in its licensing model. As AI-driven compute demand accelerates, Arm's exposure to multiple high-growth end markets positions it as a long-duration compounder rather than a cyclical semiconductor supplier.

5.4 Competitive Landscape (x86, RISC-V)

Arm competes primarily with x86 architectures, dominated by Intel and AMD, and with emerging open-source alternatives such as RISC-V. While x86 remains entrenched in legacy enterprise systems, Arm has gained traction due to superior power efficiency and growing software support.

RISC-V represents a longer-term competitive threat, particularly in cost-sensitive or sovereign technology initiatives. However, Arm's mature ecosystem, extensive software compatibility, and

established licensing relationships provide meaningful barriers to rapid displacement. The competitive balance therefore favours Arm in the medium term, particularly in AI-adjacent applications where performance-per-watt considerations are critical.

6. FORENSIC REVIEW OF THE FAILED 2020–2022 NVIDIA–ARM DEAL

6.1 Deal Timeline & Structure

In September 2020, NVIDIA announced a USD 40 billion acquisition of Arm Holdings from SoftBank. The structure combined cash and stock, reflecting NVIDIA's strong equity position during the early stages of the AI boom. The transaction aimed to combine Arm's universal CPU architecture with NVIDIA's leadership in accelerated computing, creating a vertically integrated AI compute platform.
However, from the outset, the proposal faced scrutiny due to Arm's central role as a neutral architecture provider for the global semiconductor industry. Over 1,000 companies depended on Arm's IP, raising immediate concerns about the implications of placing this infrastructure under a dominant GPU vendor.
By late 2021, regulatory resistance intensified globally, leading to prolonged investigations. NVIDIA and SoftBank ultimately abandoned the transaction in February 2022.

6.2 Global Regulatory Opposition (FTC, CMA, EU, SAMR)

The proposed acquisition triggered parallel regulatory reviews across all major jurisdictions:

US (FTC – Federal Trade Commission)

The US Federal Trade Commission filed a lawsuit to block the deal, arguing that NVIDIA would gain the ability and incentive to limit access to Arm's CPU IP for competitors in emerging markets such as autonomous driving, datacenter CPUs, and DPU architectures.

United Kingdom (CMA – Competition and Markets Authority)

The UK Competition & Markets Authority expressed two categories of concerns:

- Substantial lessening of competition because Arm's neutrality would be compromised.
- Risks to national security due to Arm's strategic importance as a UK-headquartered technology asset.

European Union

The European Commission raised issues related to reduced innovation and potential harm to downstream markets. The Commission also pointed to risks that NVIDIA could degrade Arm's open licensing model.

China (SAMR – State Administration for Market Regulation)

China's SAMR delayed approval, leveraging the review as a strategic tool amid rising US–China technology tensions. Arm's extensive licensing presence in China made China's approval critical.
The multi-jurisdictional opposition created a synchronized regulatory barrier that was unprecedented for a semiconductor acquisition.

6.3 Competitor Objections & Industry Pushback

Virtually every major semiconductor firm—including Qualcomm, Apple, Samsung, Google, and Intel—expressed objections to the deal. These companies depended on Arm's architecture, and many were direct competitors to NVIDIA in adjacent markets.

Their pushback centered on fears that NVIDIA would gain strategic visibility into competitor roadmaps, prioritize its own products, or impose restrictive licensing terms. The scale and coordination of this opposition signalled to regulators that industry-wide innovation could be impaired.

6.4 Vertical Foreclosure Concerns

The most significant regulatory argument was **vertical foreclosure**: the concern that NVIDIA would use control of Arm's CPU IP to disadvantage downstream competitors. Regulators believed that NVIDIA could:

- constrain competitor access to future Arm technology
- delay or degrade support for rival architectures
- bundle Arm IP with its own GPUs and software, increasing dependence
- foreclose the emerging cloud CPU market

Given Arm's criticality to mobile, IoT, automotive, and increasingly datacenter systems, regulators viewed these risks as systemic.

6.5 Why Remedies Failed

NVIDIA proposed behavioral remedies, including maintaining Arm's open licensing model and committing to neutrality. Regulators rejected these for three reasons:

1. **Behavioral remedies are difficult to monitor**, especially in fast-moving technology markets.
2. **NVIDIA's incentive structure** fundamentally conflicted with Arm's neutrality.
3. **Regulators doubted enforceability** over the 10–20-year horizon required for architectural stability.

In short, remedies were insufficient because the nature of the competitive harm was structural, not operational.

6.6 Final Termination in 2022 & Key Lessons

By early 2022, ongoing investigations across multiple jurisdictions made approval timelines unworkable. The deal was formally terminated in February 2022.

Key lessons include:

- Foundational IP providers cannot be acquired by dominant ecosystem players without triggering global regulatory alignment.
- Licensing neutrality is as important as market share in semiconductor evaluations.
- Geopolitics and national security now shape technology M&A decisions as much as competition analysis.

This context is essential in understanding how the 2026 landscape changes the feasibility calculus.

7. UPDATED 2026 MARKET LANDSCAPE

7.1 AI Boom & Data Center Explosion

By 2026, global demand for AI compute has accelerated far beyond 2020–22 levels. Large language models, generative AI, and cloud scale-out architectures have created unprecedented demand for accelerated computing. Training clusters and inference workloads have multiplied, with data centers becoming the fastest-growing infrastructure category globally.

This surge has increased the strategic importance of CPU efficiency, GPU–CPU coherence, and system-level optimization. Arm, once dominated by mobile, now occupies a central role in cloud compute through Arm-based server architectures.

7.2 Global Political Shifts in Semiconductors

Between 2022 and 2026, governments have significantly increased policy activity:

- The US CHIPS Act has driven domestic semiconductor expansion.
- The UK has strengthened oversight of strategic technology assets.
- China has increased support for indigenous architectures like RISC-V.
- The EU has accelerated industrial sovereignty initiatives.

However, regulators now differentiate between **horizontal mergers** (direct competitors) and **vertical integrations** that enhance supply chain resilience. Unlike 2020–22, Arm's position in 2026 is more diversified and less vulnerable to claims of total ecosystem dependency.

7.3 Licensing Models & Market Dependencies

Arm's licensing model has evolved post-IPO, with greater clarity in governance, transparency, and roadmap disclosures. Arm has introduced enhanced Chinese firewalls, third-party auditability, and formal neutrality guarantees to reassure customers.

These structural changes reduce the risk of foreclosure and weaken the central regulatory objection that blocked the 2020–22 deal.

7.4 How 2026 Differs from 2020–22

Key differences include:

- The industry is now more heterogeneous, with RISC-V rising as a credible alternative.
- Cloud providers increasingly design their own custom silicon.
- Arm's revenue dependence is more diversified, reducing fears of single-player control.
- Regulators are more focused on combating Chinese semiconductor scaling than restricting Western vertical integration.

The same deal that was unthinkable in 2022 has become strategically relevant and politically advantageous in 2026.

8. STRATEGIC RATIONALE FOR NVIDIA TO ACQUIRE ARM IN 2026

8.1 Vertical Integration Benefits

NVIDIA seeks to unify CPUs and GPUs under a single architectural roadmap, enabling deeper optimization of memory hierarchies, interconnects, compiler stacks, and power management. Control of Arm's ISA accelerates NVIDIA's ability to design full-stack AI platforms where every layer—from chip to software—is optimized for training and inference efficiency.

8.2 CPU + GPU Architecture Convergence

AI workloads increasingly require tight coupling between CPUs (task orchestration) and GPUs (parallel compute). Arm's efficiency-oriented CPU designs complement NVIDIA's GPU dominance. A unified architecture could enable:

- lower latency in AI pipelines
- greater energy efficiency
- system-level performance scaling
- integration of AI accelerators into CPU clusters

This convergence mirrors industry shifts toward heterogeneous compute standards.

8.3 Automotive, Cloud & IoT Synergies

NVIDIA's automotive division depends on integrated CPU–GPU compute. Arm's CPU architecture is already the standard for automotive systems, making integration natural.
Cloud providers (AWS, Google, Microsoft) increasingly use Arm CPUs for energy-efficient compute. NVIDIA–Arm integration could deliver purpose-built AI superchips for datacenters, offering hyperscale alternative performance paths.
In IoT and embedded markets, Arm's footprint, combined with NVIDIA's AI capabilities, positions the combined entity to dominate AI-at-the-edge.

8.4 Long-Term AI Leadership Impact

Control of Arm would elevate NVIDIA from a GPU leader to the architect of the entire global compute stack. With unified hardware, software, and licensing economics, NVIDIA could shape the next generation of AI infrastructure, influence global standards, and accelerate innovation cycles.
The strategic rationale is therefore not just synergy-driven, but existential—ensuring NVIDIA remains the central platform in the AI era.

9. VALUATION OF ARM HOLDINGS (STANDALONE)

9.1 Revenue Build-Up Model

Arm's revenue model is fundamentally different from traditional semiconductor manufacturers. Rather than relying on unit volumes of chips sold, Arm monetizes intellectual property through upfront licensing fees and long-term royalty streams. As a result, revenue growth is driven by the expansion of Arm-based silicon adoption, increased content per chip, and penetration into new compute domains rather than cyclical end-market demand alone.

The revenue forecast in this case study is constructed using a **top-down growth framework** informed by Arm's historical growth profile and forward-looking industry trends. Growth assumptions vary across scenarios to reflect uncertainty around cloud adoption, automotive AI penetration, and competitive pressures from alternative architectures.

- **Base case:** Moderate deceleration in revenue growth as Arm scales across data centers and automotive platforms.
- **Bull case:** Sustained high growth driven by accelerated adoption of Arm-based CPUs in cloud infrastructure and AI-enabled edge devices.
- **Bear case:** Slower adoption and heightened competition from x86 incumbents and RISC-V alternatives.

This structured approach allows revenue expectations to flex realistically with market conditions rather than relying on overly deterministic projections.

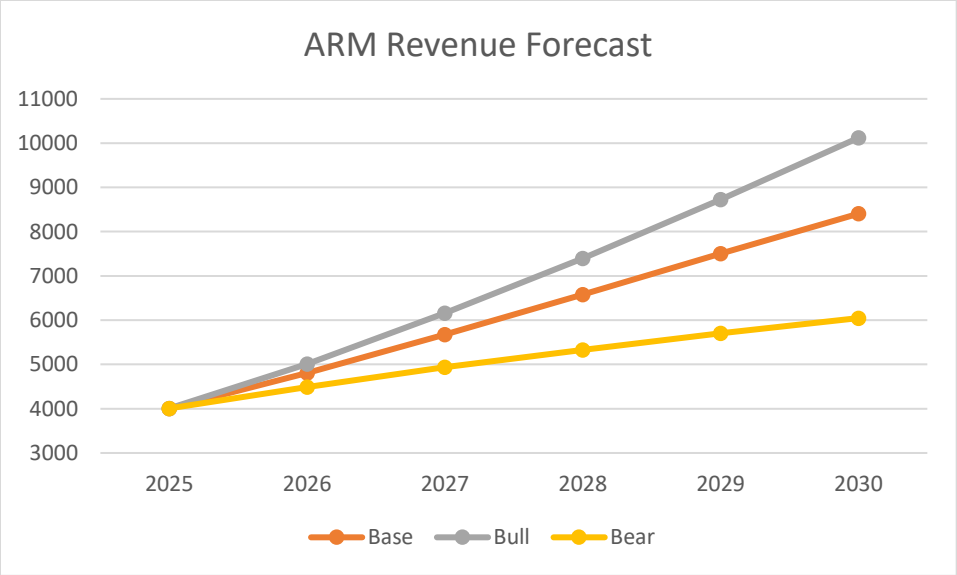

Exhibit 1: Arm Revenue Forecast by Scenario (2025–2030)

9.2 Margin Analysis & Cost Structure

Arm's cost structure reflects its position as a high-margin IP provider rather than a manufacturing-intensive company. Gross margins consistently exceed 95%, underscoring the scalability of Arm's

licensing and royalty model. Operating margins are lower due to substantial investment in research and development, which remains critical for maintaining architectural leadership and ecosystem relevance. Key margin considerations include:

- **Research & Development (R&D):** Arm's largest expense, reflecting continuous ISA evolution and CPU design enhancement.
- **Selling, General & Administrative (SG&A):** Relatively stable as a percentage of revenue, benefiting from operating leverage as revenue scales.
- **Capital Intensity:** Low capital expenditure requirements, as Arm does not own fabrication assets.

Across scenarios, operating margins expand gradually as revenue growth outpaces fixed operating costs. The bull case assumes stronger operating leverage, while the bear case reflects limited margin expansion due to competitive pricing pressure and slower ecosystem monetization.

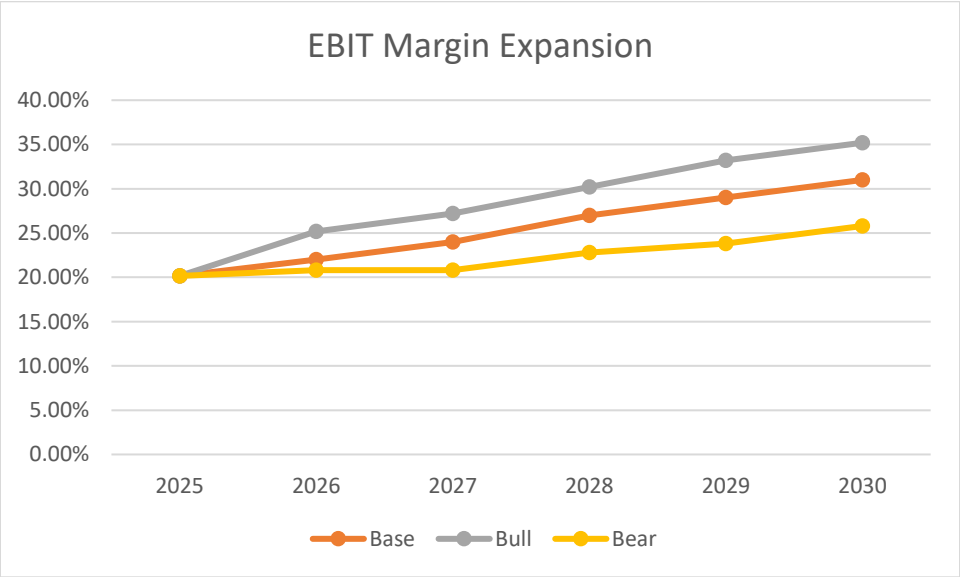

Exhibit 2: EBIT Margin Expansion (Base, Bull, Bear)

9.3 Discounted Cash Flow (DCF) Valuation

Arm's intrinsic value is estimated using a discounted cash flow methodology based on projected free cash flows and a terminal value reflecting long-term growth sustainability. Free cash flow is derived from net operating profit after tax (NOPAT), adjusted for non-cash charges, capital expenditure, and working capital requirements.

A weighted average cost of capital (WACC) is used to discount projected cash flows, capturing Arm's business risk, capital structure, and market conditions. The terminal value is calculated using a perpetual growth model, consistent with Arm's long-duration royalty economics.

In the **base case**, the DCF implies an enterprise value of approximately **USD 25–26 billion**, with terminal value accounting for the majority of total valuation. This reflects the long-term nature of Arm's licensing model, where value creation extends well beyond the explicit forecast period.

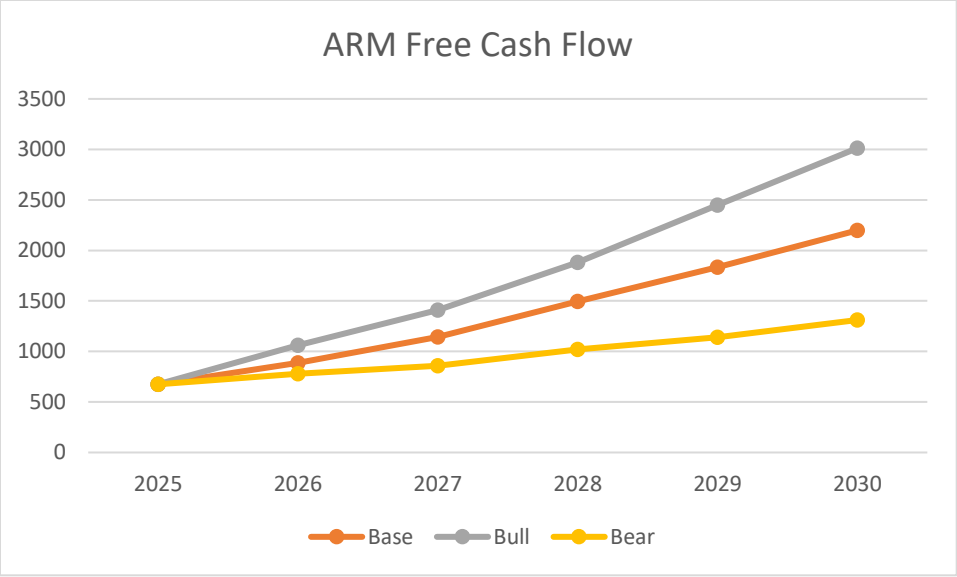

Exhibit 3: Free Cash Flow Forecast (USD millions)

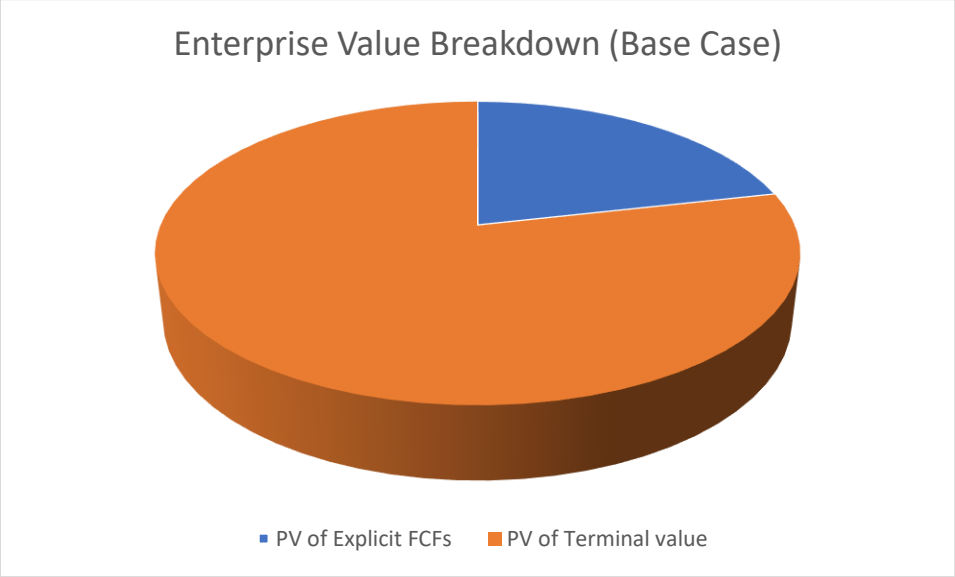

Exhibit 4: Enterprise Value Breakdown

9.4 Trading Comparables

Trading comparable analysis is used as a secondary valuation reference. Relevant peers include IP-centric semiconductor firms, high-margin platform businesses, and architecture-driven technology companies.

Compared to peers, Arm trades at a premium multiple due to:

- Structural royalty scalability
- Exceptional gross margins
- Strategic importance within AI compute ecosystems

However, valuation multiples alone fail to fully capture Arm's role as systemic infrastructure rather than a typical commercial semiconductor vendor. As such, trading comparables are used to contextualize—rather than anchor—the intrinsic valuation.

9.5 Precedent Transactions

Precedent transactions involving semiconductor IP providers and platform-level assets indicate that strategic buyers are often willing to pay significant premiums for long-term architectural control. Historical transactions suggest valuation outcomes are heavily influenced by ecosystem position and regulatory feasibility rather than short-term financial metrics.

Arm's previous proposed acquisition by NVIDIA in 2020 provides a benchmark for strategic value, although changed market conditions and regulatory frameworks necessitate a fresh valuation perspective.

9.6 Scenario Analysis (Bull, Base, Bear)

To account for uncertainty, three valuation scenarios are constructed:

- **Bull Case:** Higher revenue growth, stronger margin expansion, and a lower discount rate result in enterprise values materially above the base case, reflecting Arm's potential to become a dominant CPU architecture in AI-centric data centers.
- **Base Case:** Balanced growth and moderate margin improvement yield an enterprise value aligned with Arm's current strategic trajectory.
- **Bear Case:** Slower adoption, margin compression, and higher capital costs lead to a significantly lower valuation, reflecting downside risk from architectural competition.

Sensitivity analysis demonstrates that Arm's valuation is most sensitive to long-term growth expectations and discount rates, reinforcing the importance of ecosystem sustainability over near-term earnings volatility.

WACC \ g	2.00%	2.5%	3.00%	3.5%	4.0%
8.5%	28689.1847	30723.24	33127.11	36011.77	39537.46
9.0%	26479.9139	28191.62	30188.6	32548.69	35380.79
10.0%	22897.3773	24148.86	25579.12	28722.94	29154.79
11.0%	20119.5363	21066.2	22131.19	23338.18	24717.59
12.0%	17904.6269	18640.08	19457.24	20370.54	21397.99

Exhibit 5: Enterprise Value Sensitivity to WACC & Terminal Growth (USD Millions)

10. REGULATORY FEASIBILITY ANALYSIS (2026)

10.1 United States — FTC Assessment
In 2026, the US Federal Trade Commission remains focused on preserving competition in foundational technology markets. However, regulatory priorities have evolved toward strengthening domestic semiconductor capabilities and maintaining leadership in AI infrastructure. A renewed NVIDIA–Arm transaction would be scrutinized for foreclosure risks but evaluated within a broader national competitiveness framework.

10.2 United Kingdom — CMA Assessment
As Arm's country of origin, the UK Competition and Markets Authority would reassess the transaction through both competition and national security lenses. Structural safeguards, enforceable neutrality commitments, and continued UK-based R&D investment would likely be prerequisites for approval.

10.3 European Union — Commission Review
The European Commission's review would emphasize innovation outcomes and downstream competition. Enhanced transparency in Arm's licensing, combined with diversified ecosystem dependencies, could reduce prior concerns regarding innovation harm.

10.4 China — SAMR Considerations
China remains the most uncertain approval jurisdiction. While Arm's extensive Chinese customer base creates political sensitivity, diversification toward alternative architectures and evolving global trade dynamics may reduce China's ability to exercise unilateral veto power relative to 2022.

10.5 Probability Model & Approval Scenarios
Taking all jurisdictions into account, regulatory approval in 2026 remains challenging but more plausible than during the 2020–2022 period. The probability of approval increases materially under a scenario involving enforceable, structural remedies rather than behavioral commitments alone.

10.6 Required Remedies
Any viable transaction would require:

- Guaranteed open licensing
- Firewalled governance structures
- Independent oversight mechanisms
- Long-term commitments to architectural neutrality

These remedies would be essential to mitigate foreclosure concerns and ensure global ecosystem trust.

11. KEY RISKS & MITIGATION STRATEGIES

Despite strong strategic logic and long-term value creation potential, a renewed NVIDIA–Arm transaction would involve several material risks. These risks span technological evolution, competitive dynamics, regulatory intervention, supply chain exposure, and post-acquisition integration challenges. Each risk category is assessed below alongside potential mitigation strategies.

11.1 Technological Risks

The semiconductor industry is characterized by rapid technological change, where architectural leadership can erode quickly if innovation cycles slow. Arm faces the risk of architectural displacement if alternative instruction set architectures (ISAs), particularly RISC-V, achieve sufficient ecosystem maturity and software compatibility.

Additionally, AI workloads continue to evolve, and future compute paradigms may reduce reliance on traditional CPU orchestration, potentially limiting Arm's long-term relevance if not continuously adapted.

Mitigation:

Continued heavy investment in R&D, expansion of AI-optimized CPU architectures, and close coordination with software ecosystems are essential. Under NVIDIA ownership or partnership, Arm could accelerate architectural innovation by aligning roadmaps with next-generation AI workloads.

11.2 Competitive Risks

Arm competes against entrenched x86 incumbents (Intel and AMD) in the data center and emerging open-source alternatives such as RISC-V across IoT, automotive, and sovereign compute initiatives. Competitive pressure may constrain pricing power, slow adoption, or force higher ecosystem incentives. From NVIDIA's perspective, ownership of Arm could also provoke competitive retaliation from hyperscalers or semiconductor firms that depend on Arm's neutrality.

Mitigation:

Maintaining Arm's licensing flexibility, broad ecosystem access, and competitive pricing model is critical. Explicit governance separation and transparency measures would help preserve trust among existing licensees and prevent customer attrition.

11.3 Regulatory & Political Risks

Regulatory risk remains the single largest obstacle to any NVIDIA–Arm transaction. Authorities in the US, UK, EU, and China have previously demonstrated coordinated opposition based on concerns around vertical foreclosure, innovation harm, and national security.

Political risk is elevated due to Arm's role as foundational global infrastructure and the semiconductor industry's increasing entanglement with geopolitical strategy.

Mitigation:

Any renewed acquisition attempt would require enforceable, structural remedies rather than behavioral commitments. These include guaranteed open licensing, independent governance boards, third-party audits, and long-term neutrality covenants designed to survive political shifts.

11.4 Supply Chain Risks

Although Arm does not manufacture chips, its architecture underpins a global semiconductor supply chain that remains geographically concentrated, particularly in East Asia. Geopolitical disruptions—especially involving Taiwan—could indirectly impact Arm-based ecosystem adoption and long-term growth.

Export controls and technology restrictions could also limit access to key markets.

Mitigation:

Diversification of manufacturing partners among licensees, increased sovereign semiconductor initiatives, and regulatory cooperation frameworks reduce systemic exposure. Arm's IP-centric model inherently offers greater resilience than asset-heavy manufacturers.

11.5 Integration Risks

Post-acquisition integration poses both operational and cultural risks. NVIDIA's vertically integrated, performance-driven culture differs meaningfully from Arm's neutral, ecosystem-oriented model. Poor integration could undermine developer trust, slow innovation, or trigger customer disengagement.

Mitigation:

Preserving Arm's operational independence, retaining key engineering leadership, and implementing firewalled governance structures would be essential. Integration should prioritize strategic alignment rather than operational consolidation.

12. FINAL RECOMMENDATION

This case study concludes that Arm Holdings represents a uniquely valuable and strategically critical asset within the evolving AI compute ecosystem. Its licensing-based model, architectural ubiquity, and expanding relevance beyond mobile devices position Arm as long-duration infrastructure rather than a cyclical semiconductor company.

From a **valuation perspective**, the scenario-based DCF analysis demonstrates that Arm's intrinsic value is highly sensitive to long-term growth and discount rate assumptions. The base-case valuation supports Arm's standalone attractiveness, while the bull case highlights meaningful upside potential driven by accelerated adoption in cloud, automotive, and AI-edge computing.

From a **strategic perspective**, NVIDIA would benefit materially from deeper integration of CPU and GPU architectures, enabling system-level optimization, AI performance gains, and long-term platform leadership. Control over foundational CPU IP would strengthen NVIDIA's position across data centers, automotive systems, and emerging AI applications.

However, from a **regulatory and political standpoint**, any renewed acquisition attempt remains complex and high-risk. While the 2026 landscape differs materially from the 2020–2022 period, approval would depend on unprecedented structural remedies and sustained regulatory trust.

Final View:

Arm should be viewed as a strategic asset of exceptional long-term value. While direct acquisition by NVIDIA could unlock significant strategic benefits, such a transaction would only be viable under strict neutrality safeguards and favourable geopolitical alignment. Regardless of ownership, Arm's intrinsic value and relevance to AI compute remain robust, making strategic partnership or partial alignment viable alternatives to outright acquisition.

13. REFERENCES & APPENDIX

References

- NVIDIA Corporation Annual Reports & Investor Presentations
- Arm Holdings plc Annual Reports and IPO filings
- Regulatory filings (FTC, CMA, European Commission, SAMR)
- Industry research on semiconductor architecture and AI compute
- Public market data and comparable company disclosures

Appendix

Appendix A — Revenue Forecast by Scenario

Purpose:

This appendix summarizes Arm Holdings' projected revenue trajectory under Bear, Base, and Bull scenarios, reflecting varying assumptions around AI adoption, licensing expansion, and end-market penetration.

Table A1: Arm Revenue Forecast by Scenario (USD millions)

Year	Bear Case	Base Case	Bull Case
2025	4,007	4,007	4,007
2026	4,488	4,808	5,009
2027	4,937	5,673	6,161
2028	5,332	6,579	7,393
2029	5,705	7,505	8,724
2030	6,047	8,405	10,119

Appendix B — Operating Cost Structure & Margin Assumptions

Purpose:

This appendix outlines the key operating assumptions used to derive EBIT margins across scenarios, reflecting different outcomes in R&D efficiency, operating leverage, and cost discipline.

Table B1: Key Operating Assumptions (% of Revenue)

Item	Bear Case	Base Case	Bull Case
Cost of Sales	~3.2%	~3.0%	~2.8%
R&D Expense	49–52%	46–51%	44–50%
SG&A Expense	22–24%	20–24%	18–22%
Terminal EBIT Margin	~26%	~31%	~35%

Appendix C — Discounted Cash Flow (DCF) Valuation Summary

Purpose:

This appendix presents the standalone enterprise valuation of Arm Holdings under three discrete scenarios using a DCF methodology.

Table C1: Bear Case DCF Summary

Metric	Value
WACC	11%
Terminal Growth Rate	2%
2030 Free Cash Flow	1,311
Enterprise Value	**$12.5 billion**

Table C2: Base Case DCF Summary

Metric	Value
WACC	10%
Terminal Growth Rate	3%
2030 Free Cash Flow	2,199
Enterprise Value	**$25.6 billion**

Table C3: Bull Case DCF Summary

Metric	Value
WACC	9%
Terminal Growth Rate	4%
2030 Free Cash Flow	3,013
Enterprise Value	**$44.2 billion**

Appendix D — Sensitivity Analysis

Purpose:

This appendix evaluates valuation sensitivity to key assumptions, highlighting the variables that most materially impact Arm's enterprise value.

Table D1: EV Sensitivity to WACC and Terminal Growth (USD millions)

WACC \ g	2.00%	2.5%	3.00%	3.5%	4.0%
8.5%	28,689.18	30,723.24	33,127.11	36,011.77	39,537.46
9.0%	26,479.91	28,191.62	30,188.60	32,548.69	35,380.79
10.0%	22,897.38	24,148.86	25,579.12	28,722.94	29,154.79
11.0%	20,119.54	21,066.20	22,131.19	23,338.18	24,717.59
12.0%	17,904.63	18,640.08	19,457.24	20,370.54	21,397.99

Table D2: Enterprise Value Sensitivity to Revenue Growth & EBIT Margin (USD millions)

Rev Growth \ EBIT Margin	25.0%	27.0%	29.0%	31.0%	33.0%
12.0%	24,434.42	24,935.58	25,445.28	25,963.63	26,490.74
14.0%	24,935.58	25,446.16	25,965.29	26,493.08	27,029.64
16.0%	25,445.28	25,965.29	26,493.08	27,028.94	27,572.87
18.0%	25,963.63	26,493.08	27,028.94	27,572.87	28,124.88
20.0%	26,490.74	27,029.64	27,572.87	28,124.88	28,684.87

Appendix E — Key Assumptions Summary

Purpose:

To transparently disclose the high-level assumptions underpinning the valuation analysis.

Table E1: Valuation Assumptions Summary

Assumption	Bear	Base	Bull
Avg Revenue Growth	Low-teens	Mid-teens	Low-20s
Terminal EBIT Margin	~26%	~31%	~35%
WACC	11%	10%	9%
Terminal Growth	2%	3%	4%
CapEx (% revenue)	~2.5%	~2.5%	~2.5%
Depreciation (% revenue)	~3%	~3%	~3%